

Yiğit Sarp Avcı

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Education

Boğaziçi University - B.Sc. in Computer Engineering

09/2023 – Present

Kabataş Boys' High School

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00; Extracurriculars: Debate, Model United Nations, Entrepreneurship, and Coding Club.

Experience

Executive Board Member & DJ at Radyo Boğaziçi

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

Work and Travel Program Participant

06/2025 – 09/2025

- Gained international work experience in the USA, improving English communication, teamwork, and adaptability in a multicultural environment.

Regional Organizational Team Member at ESTIEM

09/2023 – Present

- Contributed to the planning, coordination of events, and organizational processes within ESTIEM, an international network for Industrial Engineering and Management students.

Technical Projects

AlphaTrigger AI Terminal — Python, FastAPI, Next.js, WebSockets

[GitHub Link](#)

- Autonomous high-frequency trading simulation engine featuring deterministic risk-management strategies, dynamic volatility-adjusted position sizing, and a real-time WebSocket-driven dashboard.

Performo Enterprise Admin — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

[GitHub Link](#)

- Developed an administrative dashboard with real-time operational metrics and server-side synchronized state management for system monitoring; implemented a design system with sub-second data refresh cycles via force-dynamic rendering and optimized SQL indexing for database stability.

AI-Powered FinTech Intelligence Platform — Next.js, FastAPI, PostgreSQL, Docker

[GitHub Link](#)

- Built a full-stack, deployment-ready financial dashboard combining personal finance analytics, a portfolio risk evaluator, and OpenAI-generated insights with secure NLP fallbacks; engineered a live limit-order trading simulator featuring a synchronous Price-Time Priority matching engine with automated CI/CD pipelines and high test coverage.

SchedSim CPU Scheduler — x86-64 Assembly, GNU Toolchain, Linux Syscalls

[GitHub Link](#)

- High-performance CPU scheduling simulation engine (FCFS, SJF, SRTF, PF, RR) built with zero standard library dependencies, utilizing direct Linux syscalls for I/O and process management.

Spireslayer Command Interpreter — C, Custom Backtracking Lexer

[GitHub Link](#)

- CLI-based interpreter replicating fantasy RPG REPL mechanics; implemented an autonomous backtracking lexical parser to handle non-deterministic grammar and command resolution.

Aura Functional Interpreter — Python, Functional Paradigms, Dual Scoping Engine

[GitHub Link](#)

- Interpreter for a custom functional language supporting first-class closures, higher-order function patterns, and toggleable lexical/dynamic scoping engines.

MatrixNet — Java, Tarjan's SCC Algorithm, Graph Theory

[GitHub Link](#)

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection ($O(V + E)$); optimized with Open Addressing to eliminate internal JVM string allocation overheads.

GigMatch-Pro — Java, Memory-Optimized Heaps

[GitHub Link](#)

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for priority allocation and candidate-to-job matching algorithms.

Nightpass Survival Game — Java, Rule Engine Architecture

[GitHub Link](#)

- Algorithmic implementation of a rule engine governing strategic deck, hand, and resource economics in a terminal-based survival card game.

Qt 2048 WebAssembly — C++, Qt6, WebAssembly (WASM)

[GitHub Link](#)

- Engineered a high-performance clone of the 2048 game using C++ and the Qt6 framework; cross-compiled for the web via WebAssembly (WASM) for high-fidelity browser execution with optimized grid merging logic and infinite state-history for undos.

Skills

- **Programming:** Python, Java, C/C++, Qt6, WebAssembly, TypeScript, JavaScript, SQL, x86-64 Assembly, Bash
- **Low-Level & Systems:** Linux Syscalls, Process Scheduling, Multithreading (Pthreads), Memory Management, Qt Framework
- **Enterprise Full-Stack & AI:** Next.js 15+, React, Prisma, SQLAlchemy, FastAPI, Tailwind CSS, PostgreSQL, Advanced Agentic Coding
- **Tools & Ecosystem:** Linux/Unix, Git/GitHub, Docker, Alembic, ccache, Pytest, CI/CD, Makefile, LaTeX, MS Office
- **Languages:** Turkish (native), English (fluent), German (elementary)

Volunteer & Extracurricular Experience

- **TOÇEV:** Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds. (09/2019 – 03/2020)
- **Boğaziçi University Industrial Engineering Club:** Actively contributed to the Volunteering Division. (09/2023 – 06/2024)
- **Kabataş Boys' High School CIP Club:** Organized and executed various community-oriented projects. (09/2018 – 06/2023)
- **Athletics:** Student-Athlete, Boğaziçi University Sailing Team and Ski Team & Windsurf Athlete, Competing Individually